

Sahbhagi

सहभागी — पाठकों को शिक्षित, पाठकों को प्रशिक्षित, पाठकों को

Product Requirements Document

An AI-native, multilingual audience interaction platform for trainers, educators, and government skilling programmes

Field	Detail
Document owner	Ashish
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Primary market	India - vocational training, government skilling missions, education

1. Executive Summary

Sahbhagi is an audience interaction platform - live polls, quizzes, Q&A, and surveys - built for trainers, educators, and government skilling programmes across India. It is positioned as a Slido alternative that is native to Indian languages, native to AI, and native to the offline/low-bandwidth realities of Tier 2/3 India, rather than treating these as add-ons.

The product is aimed first at vocational trainers (ITI faculty, government training partners such as MPOnline and IndiaAI Mission-aligned programmes) and rural/first-generation learners, where existing tools like Slido fall short on language coverage, bulk content creation, and affordability.

Three capabilities differentiate Sahbhagi from the incumbent and are treated as Phase 1 requirements rather than later enhancements:

- Multilingual by participant, not by event - every participant can join, answer, and be understood in their own language, including AI-assisted real-time translation.
- Excel/CSV bulk import - trainers can build an entire quiz or poll set, and a participant roster, from a spreadsheet in seconds, with round-trip export back to the same format.
- AI-native architecture - AI is a service layer available to every module (session building, moderation, translation, analytics), not a set of isolated "AI buttons."

2. Problem Statement & Opportunity

2.1 Problem

- Existing audience-interaction tools (Slido and similar) are built meeting-first and English-first; regional-language participation is a second-class experience at best.
- Trainers running the same content across many batches/trades rebuild every poll and quiz manually - there is no bulk-import path.
- AI assistance in incumbent tools is a bolt-on (a button here, a suggestion there), not something that reduces the end-to-end effort of running a session.
- Advanced features (moderation, export, branding) are gated behind pricing tiers that are unaffordable for individual trainers and small government training centres.
- Native mobile apps and offline support are weak or discontinued industry-wide, which is a hard blocker in low-connectivity training venues.

2.2 Opportunity

A leaner, AI-native, India-first product can win the vocational-training and government-skilling segment specifically, where Slido is present but not optimised - and can expand later into the same corporate/conference market Slido already serves.

3. Goals & Objectives

3.1 Business Goals

- Launch an MVP usable end-to-end (create session → run live → export results) within one training cohort cycle.
- Achieve adoption across at least one full government-partnered training programme (e.g., an ITI trade cohort or MPGB training batch) as an initial reference deployment.

- Establish multilingual + Excel import + AI-native session building as the product's defensible differentiation before competing on breadth of integrations.

3.2 Success Metrics (KPIs)

Metric	Target (first 2 cohorts)
Session completion rate (host finishes an event without abandoning)	90%
Participant join success rate (join attempt successful join)	95%
Average time to build a session via Excel import vs. manual	70% time reduction
Sessions run in a language other than English	50%
Host NPS after first session	40
AI-generated session accepted with minor edits	60%

4. User Personas

Persona	Who they are	Primary goal	Key pain point today
Host / Trainer	ITI faculty, vocational trainers, corporate facilitators	Run engaging, easy-to-build sessions	Manual content creation; language mismatch with students
Participant / Student	ITI trade students, rural/first-gen learners, meeting attendees	Participate without friction, in own language	Fear of asking "bad" questions; language barrier; low-end devices
Org Admin / Programme Owner	MPOnline/IndiaAI Mission programme managers, institution heads	Oversee usage, cost, and compliance at scale	No visibility across many trainers/batches; seat costs
Educator (recurring cohort trainer)	Faculty running the same course across many batches	Reuse content, track results per student over time	No bulk import; results not tied to real student records
Developer / Integrator	Internal MIS/LMS teams	Pull results into existing systems	No API/webhooks; no LMS sync

5. Scope

5.1 In Scope (all phases)

- Poll types: multiple choice, rating, open text, ranking, word cloud
- Quizzes with timer and leaderboard; surveys combining multiple polls
- Live Q&A with anonymous submission, upvoting, and host moderation
- Join via link, QR code, or short code; no login required for participants
- Host, Present, and Participant modes
- Multilingual UI and participant-level language selection, with AI-assisted translation
- Excel/CSV import for questions, quizzes, and participant rosters, with round-trip export
- AI-native session building, response clustering/summarisation, and post-session reporting

- Analytics dashboard, CSV/Excel/PDF export
- Organization/admin management with role-based access and bulk user management
- Branding (logo, colours) per organisation
- Integrations: PowerPoint, Google Slides, Microsoft Teams, Zoom, Webex (later phases)

5.2 Out of Scope (v1)

- Native session video recording
- Full LMS gradebook sync (tracked as a Phase 4 candidate, not committed)
- Payment/e-commerce beyond basic subscription billing
- Native iOS/Android apps at launch - mobile-first responsive web (PWA) instead

6. Functional Requirements

Each module below maps to the user story set previously agreed. Requirements are written as testable "system shall" statements grouped by module.

6.1 Event & Session Management

- The system shall allow a host to create an event with a title, and auto-generate a unique short join code and QR code.
- The system shall allow a host to add, reorder, edit, and delete interactions (polls, quizzes, Q&A, surveys) within an event before or during a live session.
- The system shall allow a host to duplicate a past event as a template for a new session.
- The system shall allow a host to password-protect an event.
- The system shall support co-host/guest moderator roles on a single event.

6.2 Poll & Interaction Types

Type	Key behaviour
Multiple choice	Single or multi-select; live bar-chart results
Rating	Numeric or star scale; average + distribution shown live
Open text	Free-text submission; each response in its own card; emoji reactions; AI clustering available
Ranking	Drag-to-order options; aggregate ranking computed live
Word cloud	Short free-text; frequency-weighted live word cloud, script-aware (Devanagari and other Indian scripts render correctly)
Quiz	Timed questions, correct-answer scoring, live leaderboard
Survey	Ordered sequence of any poll types, completed at the participant's own pace, results aggregated as one report

- The system shall allow images (upload or stock library) to be attached to any poll question.

6.3 Live Q&A & Moderation

- The system shall allow participants to submit questions anonymously by default.

- The system shall allow participants to upvote questions submitted by others, sorted by popularity or recency.
- The system shall provide a moderation queue where the host approves or rejects a question before it becomes publicly visible.
- The system shall flag likely duplicate or off-topic questions to the host using AI (see 6.9), without auto-rejecting them.

6.4 Participant Access & Join Flow

- The system shall allow participants to join via link, QR code, or short code, from any modern mobile or desktop browser, with no account or app installation required.
- The system shall let a participant select a display language at join time, independent of the language the host is using.
- The system shall function on low-end Android devices and degrade gracefully on slow (2G/3G-equivalent) connections.

6.5 Host Controls & Presentation Modes

- The system shall provide a Host mode (build and control), a Present mode (full-screen display for projection/screen-share), and the Participant mode described above, all reflecting the same live data.
- The system shall let the host switch between interactions live without disrupting connected participants.

6.6 Presentation & Meeting Integrations (Phase 2+)

- The system shall provide a PowerPoint add-in allowing polls to run natively inside a running presentation.
- The system shall provide a Google Slides add-on preserving speaker notes and slide animations while polls display inline.
- The system shall integrate as an in-meeting app for Microsoft Teams, Zoom, and Webex (Phase 2/3, prioritised by customer demand).

6.7 Multilingual (core differentiator)

- The system shall support the full host/present/participant UI in at least Hindi and English at MVP, architected to add further Indian languages without code changes to individual features.
- The system shall store language as a per-participant attribute, not a per-event attribute.
- The system shall correctly accept, store, and render text in Devanagari and other Indian scripts across open text, word cloud, and Q&A modules.
- The system shall offer AI-assisted real-time translation of Q&A/open-text submissions into the host's chosen display language.
- The system shall offer AI-assisted translation of host-authored poll content into every language selected by connected participants.

6.8 Excel/CSV Import & Export (core differentiator)

- The system shall provide a downloadable Excel template defining the expected column structure for bulk question/quiz import (question text, poll type, options, correct answer).
- The system shall validate an uploaded file before creating content, flagging missing options, ambiguous poll types, or duplicate questions, with a clear per-row error report.

- The system shall support bulk import of a participant roster (name, ID/roll number, batch/trade) and map submissions/leaderboard entries to those records.
- The system shall allow a saved question bank to be re-imported across multiple events/cohorts without re-uploading.
- The system shall export results back into the same structure as the originating import file (round-trip format), in addition to standard CSV/PDF export.

6.9 AI-Native Intelligence (core differentiator)

- The system shall generate a complete draft event (mixed poll types, quiz, Q&A framing) from a single natural-language prompt describing the session goal.
- The system shall extract poll/quiz questions automatically from an uploaded PPT, PDF, or Excel question bank.
- The system shall cluster and summarise open-text/Q&A responses into themes in real time once a configurable response threshold is reached.
- The system shall generate an automatic post-session report (key takeaways, sentiment, most-asked questions, engagement score) when a host closes an event.
- The system shall offer an optional, dismissible AI phrasing suggestion to participants before they submit a Q&A question.
- The AI capability shall be implemented as a shared service layer callable by every module (import, polls, Q&A, translation, analytics) rather than as feature-specific implementations.

6.10 Analytics & Reporting

- The system shall provide a per-event analytics view covering participation rate, response counts per interaction, and Q&A sentiment.
- The system shall support export of all results to Excel, CSV, and PDF.
- The system shall provide an organization-level analytics dashboard aggregating activity across all events for admins/owners.

6.11 Organization & Admin Management

- The system shall support role-based access control (owner, admin, host, guest moderator).
- The system shall allow bulk selection of users for role assignment or removal.
- The system shall allow an admin to enable or disable AI features org-wide.
- The system shall display seat usage and billing status to org owners/admins.

6.12 Branding & Customization

- The system shall allow an organization to apply its logo and brand colours to hosted events, available on all paid tiers (not gated to only the top tier, addressing a known competitor complaint).

6.13 Pricing & Packaging (informational - for product/business planning)

Tier	Target user	Key gates
Free	Individual trainer, trial use	Limited questions per event; no bulk import; no export
Educator / Programme	ITI faculty, single training centre	Full Excel import/export, unlimited polls, basic AI

Tier	Target user	Key gates
Institutional	Government programme (MPOnline/IndiaAI Mission partner)	Org admin tools, analytics dashboard, bulk roster import, branding
Enterprise	Corporate/conference use	Integrations (Teams/Zoom/Webex/PPT/Slides), SSO, priority support

7. Non-Functional Requirements

Category	Requirement
Performance	Live poll results shall update for all connected participants within 1 second of a new submission under normal load (500 concurrent participants per event).
Scalability	The system shall support at least 2,000 concurrent participants across all active events at MVP scale, with a horizontally scalable real-time layer.
Availability	Core live-session functionality (join, respond, present) shall target 99.5% uptime; degraded AI features must never block core polling/Q&A.
Offline / low-bandwidth	The participant experience shall function on 2G/3G-equivalent connections; the host/present experience should cache and retry silently on brief connectivity loss.
Security & privacy	Participant responses shall be anonymous by default; personal data (name, roll number) collected via roster import shall be access-controlled to the hosting organization only.
Data residency	Participant and session data for government-partnered programmes shall be stored in a manner compliant with applicable Indian data-handling requirements for public-sector engagements.
Accessibility	Participant and host UI shall meet WCAG 2.1 AA where feasible, given the first-generation-learner audience (large tap targets, minimal text density, icon + text labelling).
Localization architecture	Language shall be a first-class schema field across all modules, not a UI-only translation layer, so new features inherit multilingual support automatically.
Device support	The participant UI shall run on low-end Android browsers (Android 8) without requiring app installation.

8. System Architecture (High Level)

8.1 Component Overview

- Client layer: responsive web app (host, present, participant views) built as an installable PWA for offline-friendly participant access.
- Real-time layer: WebSocket-based pub/sub service delivering live poll/Q&A updates to all connected clients per event.
- Application/API layer: REST/GraphQL services for event, poll, Q&A, roster, and org management.

- AI service layer: a single internal service exposing session-generation, translation, clustering/summarisation, and moderation-assist capabilities, called by other modules rather than embedded per-feature.
- Import/export service: handles Excel/CSV parsing, validation, and round-trip generation, decoupled from the core poll engine so new import formats can be added independently.
- Integration layer: adapters for PowerPoint, Google Slides, Teams, Zoom, Webex, each translating the platform's native meeting/slide events into Sahbhagi API calls.
- Analytics/reporting service: aggregates event data for dashboards and export, decoupled from the real-time path so heavy queries never impact live-session performance.

8.2 AI Architecture Note

Given the zero-cost/free-tier orientation of prior work in this space, the AI service layer should be built with a pluggable model backend (a hosted LLM API called per-session, with strict per-event token/cost budgeting) rather than a self-hosted model at MVP, with the option to revisit self-hosting once usage volume justifies the infrastructure cost.

8.3 Suggested Technology Direction (indicative, not prescriptive)

- Real-time: WebSocket service (e.g. Socket.IO or a managed real-time backend) fronted by a pub/sub broker for horizontal scale.
- Backend: a typed API framework with clear module boundaries matching sections 6.1-6.13 above.
- Data store: a primary relational store for events/users/rosters/results, with a document or key-value cache for live in-session state.
- AI: hosted LLM API integration behind the internal AI service layer, with response caching for repeated translation/summarisation calls to control cost.

9. Core Data Model (Key Entities)

Entity	Key attributes
Organization	id, name, branding, tier, admin users, AI enabled flag
User	id, org_id, role, name, auth method
Event	id, org_id, host_id, title, join_code, QR, password, status, created_from (manual/AI/import)
Interaction	id, event_id, type (poll/quiz/Q&A/survey), order, content, correct_answer (quiz)
Participant	id, event_id, display_name (optional), language, roster_ref (optional), device_info
Response	id, interaction_id, participant_id, value, timestamp, upvotes, moderation_status
Roster record	id, org_id, name, external_id, batch/trade, imported_from_file_ref
Import job	id, org_id, file_ref, type (questions/roster), validation_result, status
AI job	id, event_id, type (generate/translate/cluster/summarise), input_ref, output_ref, cost/tokens
Analytics snapshot	id, event_id/org_id, metric set, generated_at

10. Third-Party Integrations

Integration	Purpose	Phase
Microsoft PowerPoint	Native in-deck polling	2
Google Slides	Native in-deck polling	2
Microsoft Teams	In-meeting app	2/3
Zoom	In-meeting polls/Q&A/quizzes	2/3
Webex	In-meeting app	3
Hosted LLM API	AI session generation, translation, clustering, summarisation	1 (basic) 3 (full)
LMS (Moodle/Canvas or India-specific equivalents)	Gradebook/result sync	4 (candidate, not committed)

11. Release Plan / Phased Roadmap

Phase	Scope
Phase 1 - MVP	Multiple choice, word cloud, open text polls; live Q&A with upvote; QR/code join; Present mode; host moderation; Hindi + English UI with participant-level language selection; basic Excel import (question bank poll/quiz) and CSV export.
Phase 2	Quizzes with leaderboard; rating polls/surveys; PowerPoint/Google Slides integration; participant roster import via Excel; round-trip export.
Phase 3	AI-native session builder (prompt full event); AI response clustering/summarisation; analytics dashboard; org/admin management; branding; AI-powered real-time translation across all supported languages.
Phase 4	Offline-first mobile PWA hardening; LMS integration (candidate); native session recording (candidate); mid-session AI interaction recommendations; additional regional languages beyond Hindi/English.

12. Risks & Assumptions

12.1 Risks

- AI hosted-API costs could scale unpredictably with usage; requires per-event budget caps from Phase 1 onward.
- Real-time infrastructure at low cost may struggle at high concurrency (large government events); needs load testing before any large-scale programme launch.
- Devanagari/Indian-script rendering in word clouds and live displays is a known technical edge case and needs early prototyping, not late-stage QA.
- Government data-handling requirements for public-sector deployments may constrain hosting/infrastructure choices and should be confirmed before Phase 3 org-admin rollout.

12.2 Assumptions

- Initial reference deployment will be within an existing government-partnered training programme (ITI/MPOnline/IndiaAI Mission-aligned), giving early real-world validation.
- Participants primarily use mobile browsers on low/mid-range Android devices; no native app is required at launch.
- English and Hindi cover the majority of the initial target audience; further language expansion is demand-driven (Phase 4).

13. Open Questions

- Which hosted LLM provider and pricing model should back the AI service layer, and what per-event token budget is acceptable at free/low-cost tiers?
- What is the minimum viable set of Indian languages beyond Hindi/English for the initial government programme rollout?
- Should org-level data for government-partnered deployments be hosted on India-based infrastructure as a hard requirement or a preference?
- What is the target pricing for the Educator/Programme tier relative to Slido's ~\$17.50/month entry point, given the affordability goal for individual trainers?

14. Appendix - User Story Backlog Reference

The full 45+ user story backlog (Host/Trainer, Participant, Org Admin, Educator, Developer, plus the Multilingual, Excel Import, and AI-Native differentiator sets) developed in the preceding product study is the authoritative backlog for sprint planning and is not duplicated here in full; each functional requirement in Section 6 traces back to one or more of those stories by module.